

Hierarchical Clustering and k Mean Clustering

Clustering is the most common form of unsupervised learning. A type of machine learning algorithm to draw inferences from unlabeled data.

Intro: group a set of data points into clusters. The goal is to have entities within a cluster be as similar as possible, but entities in one cluster be as dissimilar as possible from entities in another.

Two types:

- ① Agglomerative (bottom-up): Start with individual points and merges them.
- ② Divisive (top-down): Start with one cluster and splits it.

Before Clustering:

- 1) Scaling: To normalize the scale of feature values as each observation's feature values are represented as coordinates in n -dimensional spaces.
- 2) Missing values imputation.

Steps:

- 1) Calculating the distance and store it in a matrix.
 - Linkage Method: Complete: Maximum distance
 - Single: Minimum distance
 - Average: Mean distance
 - Centroid: Distance between cluster centroids.
- 2) Put every point in its own cluster.
- 3) Merging the closest pairs of point based on the distance matrix.
- 4) Re-calculate the distance between new and old clusters.
- 5) Repeat steps ② and ③ until only one cluster left.

Dendrograms

Visual representation of merging/splitting of clusters.
Cutting the dendrogram to get # clusters as we want.

Measuring the goodness of clusters

The most popular one is the Dunn's Index. The higher the better.

K-Means Clustering (pre-specify # clusters)

Distance used: rely on Euclidean distances.

Stability results: require a random step at its initiation that may lead to different results every time run it.

clusters: same determination as above.

Computation complexity: less expensive and can run on large datasets with reasonable time frame.

Hierarchical clustering builds a hierarchy of clusters by either merging small clusters into larger ones (agglomerative) or splitting large clusters into smaller ones (divisive). It doesn't require the number of clusters to be specified in advance and is often visualised using a dendrogram. This approach is deterministic, i.e. it produces the same result every time.

K-means clustering divides the data into a predetermined number of clusters (K). It works by assigning points to the nearest cluster centre, iteratively adjusting the centres until the clusters stabilise. It's fast, but relies on randomness for initialisation.